LLM4Rec — Homework Guidelines

Audience: Students of LLM4Rec, HSE University
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Status: Official student-facing guidelines (binding rules)

1. Course Context

This is LLM4Rec, an AI-Assisted Recommender Systems course at HSE University. You will learn both classical RecSys algorithms (matrix factorization, content-based filtering, collaborative filtering, PageRank, two-tower retrieval) and modern LLM-based approaches (LLM4Rec, conversational RecSys, agentic RecSys).

AI use is free to use. The rubric measures exactly two things about you:

```
① Understanding — c1
   Did you understand the problem? Did you find valid references?

② AI Management — c2
   Did you direct AI to produce a working solution?
   Did you verify AI's work yourself?
   Is your reasoning accurate (not just plausible)?
```

The Top-Down 5-Loop is the operational form of these two criteria:

```
① BUILD    → run a working baseline (AI may scaffold this)
② ASK WHY  → question assumptions, list alternatives (PERSONAL — your
reasoning must be accurate)
③ TEST     → verify AI outputs against theory/data/code (PERSONAL — hand-
compute at least one result)
④ EXPLAIN  → state the result in YOUR OWN WORDS (PERSONAL — not AI's summary)
⑤ IMPROVE  → iterate on one variable, repeat
```

A student who lets AI produce working code without verifying the reasoning path will often trigger Gate 0 — because AI output frequently contains subtly wrong root causes, fabricated mechanisms, or self-contradictory reasoning that look plausible but fail on verification.

See §8 for the exact rubric criteria.

2. Grading Structure

2.1 Per-Assignment (2 points max)

Each of the 10 assignments is graded on a 2-point binary framework:

Criterion	Question	Range
c1 — Problem & References	Did you identify the problem and cite valid references?	0 or 1
c2 — Solution & Evidence	Did you solve it correctly and provide verifiable evidence?	0 or 1
Total	c1 + c2	0, 1, or 2

2.2 Gate 0 (fail-fast)

A score of 0 is automatically assigned if any of the following triggers:

- c1 = 0 (major): Reference invalid (hallucinated citation, broken URL, fabricated author/year/journal, misattribution).
- c2 = 0 (major): Claim–evidence mismatch, fabricated benchmark, hallucinated API/function usage, solution doesn't actually run, evidence missing.

2.3 Scoring Decision Table

c1	c2	c=0 subtype	Final score
1	1	(both clean)	2
1	0	c2 = minor	1
0	1	c1 = minor	1
0	0	both minor	0 (fail-fast)
1	0	c2 = major	0 (Gate 0)
0	1	c1 = major	0 (Gate 0)
0	0	any major	0 (Gate 0)

Plain English: Two clean criteria → 2 points. One minor issue → 1 point. Anything major (invalid reference or fabricated evidence) → 0 points, full stop.

2.4 Overall (10 assignments × 2 = 20 points max)

```
Total = Σ (score per assignment) ∈ [0, 20]
```

Suggested mapping (instructor reserves the right to adjust):

Total	Suggested grade band
18–20	A — Excellent
14–17	B — Good
10–13	C — Satisfactory
6–9	D — Needs improvement
0–5	F — Fail

3. The 10 Assignments (One per Week)

#	Week	Topic	Length limit
A01	W1	Random Lunch Generator — web app scaffold	≤5 pages (excl. references)
A02	W2	Content-Based Filtering implementation	≤5 pages (excl. references)
A03	W3	Collaborative Filtering implementation	≤5 pages (excl. references)
A04	W4	Association Rule mining (Apriori / FP-Growth)	≤5 pages (excl. references)
A05	W5	PageRank-based RecSys	≤5 pages (excl. references)
A06	W6	Two-Tower Retrieval (item/user embedding)	≤5 pages (excl. references)
A07	W7	Context Encoder (DL-based RecSys)	≤5 pages (excl. references)
A08	W8	Personalized Ranker	≤5 pages (excl. references)
A09	W9	LLM4Rec baseline	≤5 pages (excl. references)
A10	W10	Capstone preparation (extended scope)	≤5 pages (excl. references)

Page limit: ≤5 pages (excl. references) — hard cap. Earlier assignments may be shorter; later assignments should approach the 5-page limit — these are the rehearsal for the final capstone short paper in W12.

The instructor may revise topics and ordering. You will receive a per-assignment specification sheet at the start of each week.

4. Submission Requirements

4.1 Required files (2 files per assignment)

File	Format	Required
report.pdf	PDF	Yes
session.json	JSON (tool-native)	Yes

Note: Earlier drafts of these guidelines mentioned DOCX format. As of v1.0, the official submission format is PDF. Convert your DOCX (or markdown) to PDF before submission.

report.pdf: Your homework report, structured as described in §5.

session.json: The full session log from your AI coding tool (OpenCode, Claude Code, Cursor, or equivalent). Tool-native format is accepted. The session must show the AI tool calls, file modifications, and key milestones. This is the audit trail for your AI-assisted work.

4.2 File naming convention

```
<student-id-or-name>_<assignment-id>_report.pdf
<student-id-or-name>_<assignment-id>_session.json
```

Examples:

- jin_seungmin_a01_report.pdf
- jin_seungmin_a01_session.json
- team_alpha_a03_report.pdf
- team_alpha_a03_session.json

4.3 Deadline

Each assignment has a strict deadline announced at the start of the week. Submissions are accepted up to and including the deadline timestamp.

4.4 Late submissions

Late submissions are NOT accepted. No exceptions.

If you anticipate missing a deadline, discuss it with the instructor before the deadline, not after. Late submissions will not be graded regardless of the reason (technical issues, illness, etc. — claim exceptions in advance).

4.5 Where to submit

Submission channel: announced in class and via LMS official channel.

5. Report Structure (IEEE-aligned)

Your report.pdf must follow this structure. Templates are provided separately; this section specifies the required content.

Title block (top)

```
[Concise paper-style title, e.g., "Matrix Factorization with Side Information
on MovieLens-1M: A Lightweight Reproducibility Study"]

Student: [Name]    Team: [Team or "Individual"]
Email: [email]      Date: YYYY-MM-DD
Assignment: [Topic] (Week [N])
```

Page limit

Maximum 6 pages, excluding references. This is a hard cap.

- Earlier assignments (W1–W2) typically need only 1–2 pages — that's fine, but no more than 5.

- Later assignments (W7–W10) should approach the 6-page limit; this is good practice for the capstone short paper.
- References are NOT counted toward the 6-page limit.

Abstract

- Length: 50–150 words.
- Structure: 4 sentences — (1) Problem, (2) Method, (3) Key Result with one specific number, (4) One takeaway.

Index Terms (optional)

3–5 keywords in IEEE format: Index Terms — keyword1, keyword2, keyword3, keyword4, keyword5.

§1. Introduction

- Problem statement (2–4 sentences)
- Motivation (2–3 sentences)
- Concrete example (1 prompt / scenario / input-output pair)
- Contributions (1–2 bullets, no over-claim)

§2. Related Work

Prior work consulted: 3–7 entries in IEEE numbered format.

```
[N] A. Surname, "Title of paper," Journal/Conf., vol. x, no. x, pp. xx–xx, Year. [Online]. Available: URL. [Accessed: YYYY-MM-DD].
```

Alternatives considered: 1–3 brief mentions + 1-line rationale each.

§3. Method

- Approach (2–4 sentences)
- Pipeline (diagram or step-by-step)
- Tools & libraries (bullet list with versions, reference official docs)
- Key design decisions (2–3 non-obvious choices + rationale)
- Configuration (hyperparameters, seeds, environment)

§4. Experiments

- Setup (dataset, eval protocol, hardware)
- Results (table or figure, each row/column labeled)
- Comparison vs baseline (2–3 sentences)
- Verification (sanity checks, unit tests, manual inspection — cite specific check)

§5. Discussion

This is the most heavily weighted section for critical-thinking credit. Include:

- Failure case (1 realistic failure — error message, log excerpt)
- Root cause (specific explanation)
- Fix + verification (detection → change → confirmation)
- What worked (2–3 bullets)
- What surprised you or didn't work (2–3 bullets)
- Next improvement (1 concrete improvement with more time)

§6. AI Usage Disclosure

- AI tools used (names + versions)
- How AI was used (categorize: search, code generation, debugging, docs, viz)
- What I personally verified (for each AI output: how you confirmed correctness)
- What I trusted without verification (honest disclosure — be honest, this won't hurt you if you're transparent)
- Session log reference (the attached session.json is the audit trail)

References

IEEE numbered format, continuing from §2 if needed.

6. Class Activity (in-class presentations)

6.1 Format

Each class session, 10 students will be selected to present their work from the previous assignment. The selection is made by the instructor and is not negotiable.

After each presentation, the class engages in structured discussion. Other students ask questions, challenge assumptions, suggest alternatives.

6.2 What we grade

- Presentation quality when you are selected (clarity, evidence, response to questions)
- Active participation in discussion (asking substantive questions, offering constructive critique, building on others' work)
- Constructive feedback to peers

6.3 What this is and is not

- This IS: An opportunity to practice the Top-Down ⑤ IMPROVE step in front of others; a forcing function for deep engagement with your own work.
- This is NOT: A popularity contest. Quality of engagement matters more than quantity of comments.

6.4 Connection to grading

Class activity is 10% of the final course grade. It is graded per-session and aggregated. Missing class without prior arrangement reduces your class activity grade. Being unprepared when selected to present (e.g., "I didn't do the homework this week") is a serious negative signal.

7. The Top-Down 5-Loop — Pedagogical Core

This course is built around a specific learning philosophy. Read this section carefully; it explains why the homework is structured the way it is.

7.1 The 5 steps

```
① BUILD    → run a working baseline (you can use AI to scaffold this)
② ASK WHY  → question assumptions, list alternatives (PERSONAL)
③ TEST     → verify AI outputs against theory/data/code (PERSONAL)
④ EXPLAIN  → state the result in YOUR OWN WORDS (PERSONAL)
⑤ IMPROVE  → iterate on one variable, repeat
```

7.2 The trap we are avoiding

It is tempting to:

- Prompt AI: "Build me a RecSys with X"
- AI returns working code + explanation
- Submit the AI's output

This is the AI-delegation trap. Students who do this:

- Score 0 on §5 Discussion (it reads like AI output)
- Score 0 on §6 AI Usage Disclosure (claim verified what wasn't)
- Often trigger Gate 0 (claims unverified)
- Do not actually learn the material

7.3 What we expect instead

- Prompt AI to scaffold (step ①)
- Stop and ask: "Is this the right approach? What are the alternatives? Why this and not that?" (step ②)
- Manually verify at least one number, one derivation, one assumption (step ③)
- Write the discussion in your own voice — not AI's summary (step ④)
- Identify what to do differently next based on what you learned (step ⑤)

Students who engage with all 5 steps score 2/2. Students who only do step ① score 0–1/2.

7.4 The pedagogical quote

The course material includes this quote (from Page 5 of the introductory slides):

"AI does not do the learning for you. It shortens the loop through which you learn."

This is not a metaphor. It is the operational principle of this course.

8. Rubric Criteria in Detail

8.0 Two things we measure

```
c1 — Understanding: problem + references
c2 — AI Management: solution + reasoning + verification
```

Both binary (0 or 1). Subtypes (clean / minor / major) capture how the criterion failed.

8.1 c1 — Understanding

Score	Subtype	What triggered it
1	Clean	Clear problem statement, all references valid, attribution accurate
0	Minor	Reference format typo, outdated reference, 1–2 missing references
0	Major (Gate 0 trigger)	Hallucinated citation, broken URL, fabricated author/year, misattribution, problem misidentified

8.2 c2 — AI Management

c2 = 1 requires three aspects to be yes:

Aspect	Pass condition
Solution works	Code runs; chart/table matches code output
Reasoning accurate	Each loop step is factually correct AND logically valid
Verification cited	§6 lists at least one of: hand-computed metric, re-ran experiment, paper read beyond abstract, cross-checked number against code, argued derivation mathematically

Subtype breakdown:

Subtype	Trigger
Clean (c2=1)	All three aspects pass
Minor	One aspect weak: verification missing; OR partial evidence; OR slight over-claim; OR failure case shallow
Major (Gate 0)	Solution broken (doesn't run, fabricated benchmark, hallucinated API); OR reasoning broken (wrong root cause, wrong alternative rejection, hallucinated mechanism); OR critical

	fabrication (claimed verification without doing it)

What "accurate reasoning" means in practice:

Step	Accurate reasoning	Inaccurate reasoning
② **ASK WHY**	Adam adapts learning rate per parameter, useful for sparse RecSys embeddings	Adam is always better than SGD
③ **TEST**	Hand-computed P@10 for user 1 = 3/10 = 0.3, matches the chart value	The AI said it's right
④ **EXPLAIN**	Model learned to recommend popular items because ratings skew toward common products	It works because AI
⑤ **IMPROVE**	I'd add L2 regularization next — current model overfits per the train/test gap in Fig. 2	Make it better

What counts as "verified":

To pass the verification aspect, do at least ONE of the following for a critical output:

- Hand-computed a metric on at least one data point (show your work)
- Re-ran the experiment from scratch and confirmed identical results
- Read the cited paper beyond the abstract (or the cited source's key section)
- Cross-checked the claimed number against the actual code output (line:line)
- Argued mathematically why the formula is correct, not just "the AI said so"

If you cannot do any of the above for a critical output, you did not verify it. Write that honestly in §6.

9. AI Usage Policy

9.1 Allowed

- Using any AI coding tool (OpenCode, Claude Code, Cursor, Codex, etc.)
- Using AI to scaffold code, suggest approaches, search references, debug
- Using AI to draft text, then rewriting in your own voice
- Using AI for visualization spec generation (Plotly, D2)

9.2 Required

- Disclosure: §6 must list every AI tool and what it was used for
- Verification: Critical outputs (algorithms, math, benchmark numbers) MUST be verified
- Honesty: "Trusted without verification" is acceptable IF you're transparent

9.3 Not allowed

- Submitting AI output verbatim without personal engagement with steps ②–⑤

- Fabricating AI usage: claiming you used AI when you didn't, or vice versa
- Hallucinated citations: copying AI-suggested references without checking they exist
- Claiming verification you didn't do: this is the most common Gate 0 trigger

9.4 The session.json audit trail

The session.json you submit is the primary verification tool the instructor uses. It records:

- Every prompt you sent to the AI
- Every tool the AI invoked
- Every file the AI created or modified
- Timestamps and session duration

If your §6 claims don't match your session.json (e.g., you claim "5 hours of AI-assisted work" but session.json shows 12 minutes), this is a contradiction and may trigger Gate 0.

10. Failure Modes (What Triggers Gate 0)

Failure	Section affected	Gate 0?
Hallucinated citation (paper doesn't exist)	§2 Related Work	Yes
Fabricated benchmark number (chart shows 0.187 but code outputs 0.083)	§4 Experiments	Yes
Solution claimed to work but doesn't run	§4 Experiments	Yes
Cherry-picked metric (only positive results shown)	§4 Experiments	Yes
AI usage claimed but session.json missing	§6 AI Usage	Yes (over-claim)
Reference with broken URL (404)	§2 Related Work	Yes
Hallucinated API call (function doesn't exist or not used)	§3 Method	Yes
Discussion section is 3 generic bullets	§5 Discussion	Yes (thin 5-Loop)

11. Worked Examples (How the Rubric Plays Out)

11.1 Example 1: Score 2/2

A student submits a report with:

- All references valid (verified by instructor)
- Results table with numbers matching code output
- Failure case with root cause and fix

- Discussion section with 5-Loop reflection (5.1–5.5 all populated)
- session.json shows 4 hours of substantive work

Score: c1 = 1, c2 = 1 → 2 points.

11.2 Example 2: Score 0/2 (Gate 0)

A student submits a report where:

- §2 cites paper [3] that doesn't exist (DOI 404)
- Everything else looks great

Score: c1 = 0 (major, hallucinated citation) → Gate 0 → 0 points regardless of c2 quality.

11.3 Example 3: Score 1/2

A student submits a report with:

- All references valid
- Solution works but evidence is just one screenshot (no execution log)
- Failure case mentioned but root cause not analyzed

Score: c1 = 1, c2 = 0 (minor) → 1 point.

12. Frequently Asked Questions

Q: I used no AI at all. Do I still need §6?

A: Yes. §6 should state "No AI tools were used for this assignment." This is honest disclosure.

Q: Can I submit a Word document?

A: No. The submission format is PDF. Convert your DOCX/markdown to PDF before submitting.

Q: My session.json is huge (>10 MB). What do I do?

A: Contact the instructor before submission. We may accept gzipped or aggregated logs.

Q: Can I edit my session.json?

A: No. The session.json must be the unmodified export from your AI tool. Modifications compromise audit integrity.

Q: I missed the deadline by 5 minutes. Is that "late"?

A: Yes. Strict deadline. No exceptions.

Q: What's the page limit for the report?

A: Maximum 6 pages excluding references. References are listed separately and not counted. Earlier assignments may be shorter; later assignments should approach 6 pages.

Q: What if my AI tool crashed mid-session?

A: Document it in §6 (AI Usage Disclosure → "Session Notes"). Submit whatever session.json was produced. If critical, contact instructor in advance.

Q: Do I have to present in every class?

A: No. 10 students per class are selected. Over the course, the selection covers everyone (roughly).

Q: I'm presenting but my work has a bug. Should I hide it?

A: Absolutely not. §5 Discussion exists exactly for this. "I had a bug, root cause was X, I fixed it by Y" is exactly what we want to see.

— end of guidelines —